

Project Report: Customer Segmentation for a Retail Company

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1. Introduction

Customer segmentation is a crucial application of unsupervised machine learning in the retail industry. It involves dividing a company's customer base into distinct groups (clusters) based on shared characteristics, such as age, gender, annual income, and spending behavior¹.

By clustering customers, businesses can move away from a "one-size-fits-all" marketing strategy. Instead, they can target specific groups with tailored products and marketing campaigns, ultimately improving customer satisfaction and increasing revenue.

This project utilizes **K-Means Clustering**, a popular algorithm that partitions data into k distinct clusters, ensuring that data points within a group are similar while being different from those in other groups².

2. Objective

The primary objectives of this project are:

1. **To implement the K-Means Clustering algorithm** using Python to segment retail customers.
2. **To determine the optimal number of customer segments** using the Elbow Method.
3. **To visualize and analyze the clusters** based on two key metrics: *Annual Income* and *Spending Score*.
4. **To provide actionable insights** that can help the retail company understand its customer demographics better.

3. Methodology

3.1 Tools and Technologies

The project is implemented using the **Python** programming language with the following libraries:

- **Pandas & NumPy:** For data generation, manipulation, and analysis.
- **Matplotlib & Seaborn:** For data visualization (Elbow graph and Cluster scatter plots)³.
- **Scikit-learn:** For data preprocessing (StandardScaler) and implementing the K-Means algorithm⁴⁴⁴⁴.

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3.2 Implementation Steps

1. **Data Generation:** Since a real dataset was not provided, a dummy dataset was generated programmatically containing fields for CustomerID, Gender, Age, Annual Income, and Spending Score.
2. **Data Preprocessing:**
 - The features selected for clustering were Annual Income (k\$) and Spending Score (1-100).
 - **Feature Scaling:** The data was standardized using StandardScaler to ensure that income (which has large values) does not dominate the distance calculations compared to the spending score⁵.
3. **Finding Optimal Clusters (Elbow Method):**
 - The Within-Cluster Sum of Squares (WCSS) was calculated for cluster counts ranging from 1 to 10.
 - A graph of WCSS vs. Number of Clusters was plotted to identify the "elbow point"⁶.
4. **Model Training:** The K-Means algorithm was applied with the optimal number of clusters ($k=5$).
5. **Visualization:** The resulting clusters were plotted on a 2D scatter plot to observe the segmentation.

4. Code and Implementation Details

The core logic of the project is implemented in Python. Below are the key snippets explaining the functionality.

A. Data Preprocessing and Scaling

We focused on Income and Spending Score for the segmentation. Scaling was applied to normalize the data distribution.

```
features = ['Annual Income (k$)', 'Spending Score (1-100)']  
X = df[features]
```

```
# Standardize the features  
scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X)
```

B. The Elbow Method We used a loop to fit the K-Means algorithm for 1 to 10 clusters and stored the inertia (WCSS) to determine the best \$k\$.

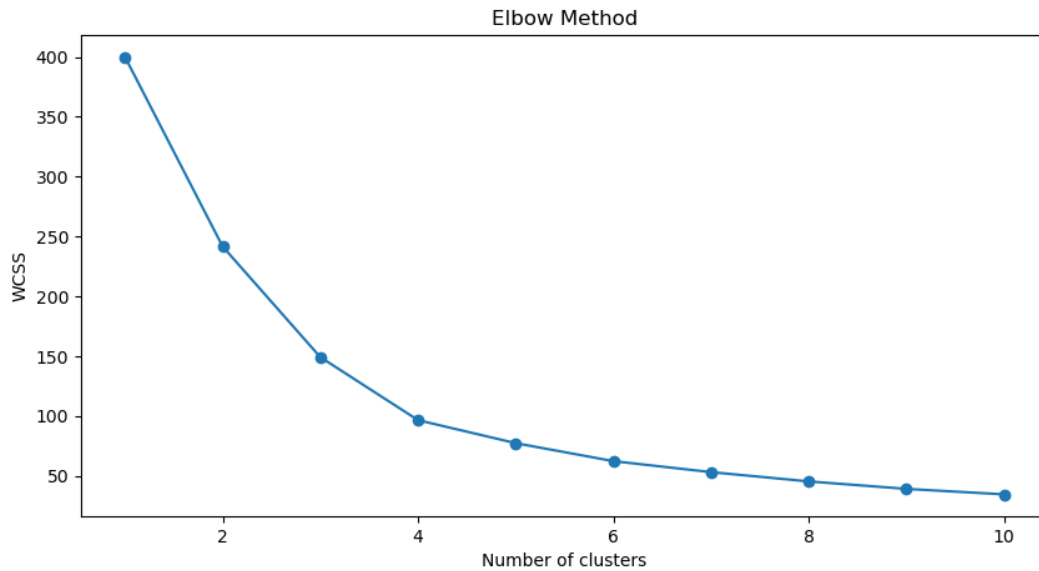
```
wcss = []  
for i in range(1, 11):  
    kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init=10, random_state=42)  
    kmeans.fit(X_scaled)  
    wcss.append(kmeans.inertia_)
```

C. K-Means Model Training Based on the Elbow graph, 5 clusters were chosen. The model assigns a cluster index (0-4) to every customer.

```
optimal_clusters = 5  
kmeans = KMeans(n_clusters=optimal_clusters, init='k-means++', max_iter=300, n_init=10,  
random_state=42)  
clusters = kmeans.fit_predict(X_scaled)  
df['Cluster'] = clusters
```

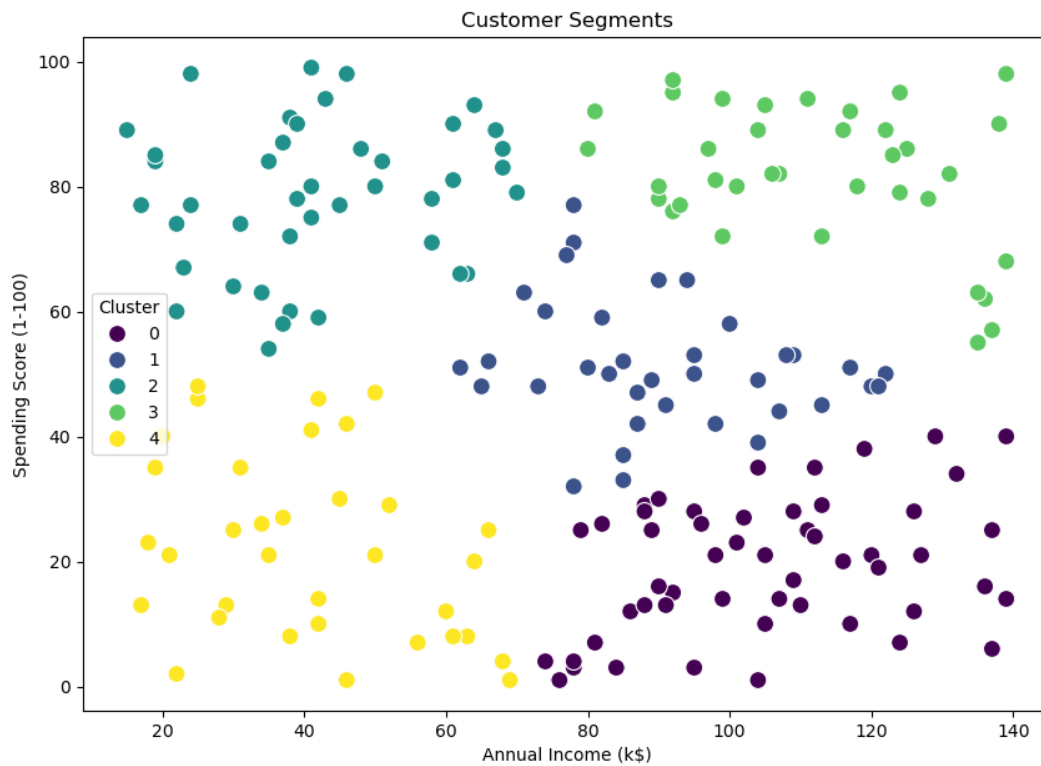
5. Results and Observations

5.1 The Elbow Method Graph



Observation: The WCSS decreases as the number of clusters increases. The graph shows a significant "bend" or "elbow" at $k=5$. This indicates that 5 is the optimal number of clusters to model this data without overfitting.

5.2 Customer Segments Visualization



Observation: The scatter plot reveals 5 distinct customer groups:

1. **Low Income, Low Spending:** Customers who earn less and spend less.
2. **Low Income, High Spending:** Customers with lower income but high spending habits.
3. **High Income, Low Spending:** High earners who are frugal.
4. **High Income, High Spending:** The target group for premium products (high value).
5. **Average Income, Average Spending:** The standard customer base falling in the middle.

6. Conclusion

In this project, we successfully implemented a **Customer Segmentation** model using the K-Means clustering algorithm.

Key Learnings:

- **Unsupervised Learning:** We learned how to find patterns in unlabelled data.
- **Data Scaling:** We observed that scaling features is essential for distance-based algorithms like K-Means to perform accurately.
- **Business Value:** The segmentation provided clear categories (e.g., "High Income-High Spending"), which allows the retail company to devise specific strategies—such as offering loyalty programs to high spenders or budget deals to low-income groups.

This project demonstrates how AI can transform raw data into actionable business intelligence.